

Table 1. Full MGSM multilingual-transfer results for Qwen3-4B, Llama-3.2-3B-Instruct, and Phi-4-mini-instruct on 100 matched questions per language. Values are full-sample means $\pm$ ten-group matched-question jackknife standard errors. Higher is better for every column.

Model	Language	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Chinese	Original	71.0 $\pm$ 4.1	0.00 $\pm$ 0.00	1.80 $\pm$ 0.05	1.71 $\pm$ 0.05
		A-LQR	20.0 $\pm$ 2.6	1.98 $\pm$ 0.02	1.53 $\pm$ 0.03	1.15 $\pm$ 0.03
		H_∞ (ours)	51.0 $\pm$ 3.5	2.00 $\pm$ 0.00	1.87 $\pm$ 0.05	1.52 $\pm$ 0.04
	French	Original	60.0 $\pm$ 4.5	0.00 $\pm$ 0.00	1.93 $\pm$ 0.03	1.70 $\pm$ 0.04
		A-LQR	29.0 $\pm$ 5.0	1.94 $\pm$ 0.04	1.62 $\pm$ 0.05	1.11 $\pm$ 0.03
		H_∞ (ours)	48.0 $\pm$ 4.7	1.96 $\pm$ 0.04	1.96 $\pm$ 0.02	1.42 $\pm$ 0.07
	Japanese	Original	56.0 $\pm$ 4.8	0.00 $\pm$ 0.00	1.80 $\pm$ 0.04	1.64 $\pm$ 0.04
		A-LQR	32.0 $\pm$ 2.5	1.84 $\pm$ 0.06	1.63 $\pm$ 0.06	1.01 $\pm$ 0.03
		H_∞ (ours)	40.0 $\pm$ 4.9	1.98 $\pm$ 0.02	1.71 $\pm$ 0.07	1.20 $\pm$ 0.05
	Swahili	Original	15.0 $\pm$ 4.3	0.00 $\pm$ 0.00	1.43 $\pm$ 0.07	1.23 $\pm$ 0.05
		A-LQR	12.0 $\pm$ 3.6	1.40 $\pm$ 0.09	1.32 $\pm$ 0.07	0.97 $\pm$ 0.03
		H_∞ (ours)	14.0 $\pm$ 3.7	0.92 $\pm$ 0.12	1.44 $\pm$ 0.07	1.06 $\pm$ 0.04
	Telugu	Original	6.0 $\pm$ 2.7	0.00 $\pm$ 0.00	1.83 $\pm$ 0.04	1.05 $\pm$ 0.02
		A-LQR	19.0 $\pm$ 2.8	1.88 $\pm$ 0.04	1.46 $\pm$ 0.06	1.18 $\pm$ 0.03
		H_∞ (ours)	8.0 $\pm$ 2.5	0.34 $\pm$ 0.08	1.33 $\pm$ 0.04	1.05 $\pm$ 0.03
Llama-3.2-3B	Chinese	Original	43.0 $\pm$ 3.3	0.00 $\pm$ 0.00	1.79 $\pm$ 0.07	1.59 $\pm$ 0.07
		A-LQR	26.0 $\pm$ 3.7	1.80 $\pm$ 0.04	1.49 $\pm$ 0.08	1.21 $\pm$ 0.08
		H_∞ (ours)	22.0 $\pm$ 4.7	1.98 $\pm$ 0.02	1.46 $\pm$ 0.06	1.35 $\pm$ 0.05
	French	Original	49.0 $\pm$ 4.6	0.00 $\pm$ 0.00	1.83 $\pm$ 0.03	1.71 $\pm$ 0.04
		A-LQR	43.0 $\pm$ 4.2	1.24 $\pm$ 0.09	1.76 $\pm$ 0.04	1.09 $\pm$ 0.03
		H_∞ (ours)	6.0 $\pm$ 2.7	0.66 $\pm$ 0.06	0.53 $\pm$ 0.04	0.47 $\pm$ 0.04
	Japanese	Original	27.0 $\pm$ 3.0	0.00 $\pm$ 0.00	1.56 $\pm$ 0.08	1.34 $\pm$ 0.09
		A-LQR	26.0 $\pm$ 3.7	1.94 $\pm$ 0.04	1.68 $\pm$ 0.08	1.31 $\pm$ 0.07
		H_∞ (ours)	12.0 $\pm$ 3.3	1.98 $\pm$ 0.02	1.31 $\pm$ 0.10	1.16 $\pm$ 0.08
	Swahili	Original	40.0 $\pm$ 3.3	0.00 $\pm$ 0.00	1.79 $\pm$ 0.02	1.49 $\pm$ 0.03
		A-LQR	22.0 $\pm$ 4.4	0.68 $\pm$ 0.10	1.47 $\pm$ 0.04	0.98 $\pm$ 0.04
		H_∞ (ours)	14.0 $\pm$ 4.0	1.90 $\pm$ 0.04	1.28 $\pm$ 0.06	1.05 $\pm$ 0.05
	Telugu	Original	6.0 $\pm$ 2.2	0.00 $\pm$ 0.00	1.43 $\pm$ 0.04	1.07 $\pm$ 0.03
		A-LQR	21.0 $\pm$ 6.0	1.86 $\pm$ 0.05	1.44 $\pm$ 0.08	1.25 $\pm$ 0.07
		H_∞ (ours)	7.0 $\pm$ 2.6	0.82 $\pm$ 0.06	0.53 $\pm$ 0.10	0.54 $\pm$ 0.09
Phi-4-mini	Chinese	Original	63.0 $\pm$ 5.2	0.00 $\pm$ 0.00	1.90 $\pm$ 0.04	1.80 $\pm$ 0.04
		A-LQR	0.0 $\pm$ 0.0	1.10 $\pm$ 0.10	0.05 $\pm$ 0.03	0.08 $\pm$ 0.03
		H_∞ (ours)	47.0 $\pm$ 4.2	1.64 $\pm$ 0.09	1.72 $\pm$ 0.05	1.39 $\pm$ 0.07
	French	Original	58.0 $\pm$ 3.6	0.00 $\pm$ 0.00	1.91 $\pm$ 0.02	1.64 $\pm$ 0.05
		A-LQR	0.0 $\pm$ 0.0	1.10 $\pm$ 0.08	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		H_∞ (ours)	49.0 $\pm$ 5.0	1.40 $\pm$ 0.05	1.90 $\pm$ 0.03	0.98 $\pm$ 0.03
	Japanese	Original	49.0 $\pm$ 3.8	0.00 $\pm$ 0.00	1.91 $\pm$ 0.03	1.57 $\pm$ 0.04
		A-LQR	0.0 $\pm$ 0.0	1.74 $\pm$ 0.08	0.00 $\pm$ 0.00	0.03 $\pm$ 0.02
		H_∞ (ours)	41.0 $\pm$ 4.1	0.62 $\pm$ 0.10	1.68 $\pm$ 0.06	1.20 $\pm$ 0.05
	Swahili	Original	32.0 $\pm$ 3.9	0.00 $\pm$ 0.00	1.67 $\pm$ 0.05	1.36 $\pm$ 0.08
		A-LQR	0.0 $\pm$ 0.0	1.56 $\pm$ 0.09	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		H_∞ (ours)	21.0 $\pm$ 2.8	1.00 $\pm$ 0.14	1.18 $\pm$ 0.09	0.77 $\pm$ 0.08
	Telugu	Original	25.0 $\pm$ 3.7	0.00 $\pm$ 0.00	1.57 $\pm$ 0.04	1.25 $\pm$ 0.05
		A-LQR	0.0 $\pm$ 0.0	1.84 $\pm$ 0.04	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
		H_∞ (ours)	18.0 $\pm$ 2.0	1.38 $\pm$ 0.10	1.34 $\pm$ 0.04	0.89 $\pm$ 0.07